

# *Ball Valves*



ASME Class 150 to 2500 | 8 mm - 200 mm (1/4" - 8")  
ISO 17292/ API 6D

L&T Valves Limited (Formerly Audco India Limited) is a wholly owned subsidiary of Larsen & Toubro. Backed by a fifty-year track-record of excellence and world-leading innovation, the company provides engineered flow-control solutions for key sectors of the economy such as oil & gas, power, petrochemicals, chemicals, fertilizers and pharmaceuticals.

**Product Range:**

- Gate, Globe & Check Valves
- Valves for Power
- Pipeline & Process Ball Valves
- Triple-offset Butterfly Valves
- Flanged & Wafer-type Butterfly Valves
- Double Block and Bleed Plug Valves
- Control Valves
- Customised Solutions

Designs for the valves are created by an experienced team of valve experts who have a deep understanding of user-industry processes. An extensive manufacturing and quality assurance infrastructure ensure that world-class designs are transformed into high performance products. Every phase of manufacture is governed by an institutionalised environment, health and safety policy.

L&T Valves marketing network spans the globe reinforced by strategic alliance with key international distributors. In India, the company has a presence in every industrial centre through a network of offices, field engineers, distributors, automation centres and service franchisees.



- Single-piece Range - Page No. 9
- Two-piece Range - Page No. 10
- Three-piece Range - Page No. 12



L&T Valves manufactures a comprehensive range of Ball Valves in sizes up to 8" (DN 200) and in ASME classes from 150 to 2500. The valves are offered in combination of size, pressure class, material of construction, end-connection, etc., to suit myriad applications in process plants, utility lines and HVAC.

### Product Portfolio

| Valve                      | Type      | End Connection           | 1/4 | 3/8 | 1/2 | 3/4 | 1  | 1 1/4 | 1 1/2 | 2  | 2 1/2 | 3  | 4   | 6   | 8   |
|----------------------------|-----------|--------------------------|-----|-----|-----|-----|----|-------|-------|----|-------|----|-----|-----|-----|
|                            |           |                          | 8   | 10  | 15  | 20  | 25 | 32    | 40    | 50 | 65    | 80 | 100 | 150 | 200 |
| Single-piece, Regular Bore | Fire-safe | Flanged CI 150/ 300      |     |     | •   | •   | •  |       | •     | •  | •     | •  | •   | •   |     |
| Two-piece, Full Bore       | Fire-safe | Flanged CI 150/ 300      |     |     | •   | •   | •  |       | •     | •  | •     | •  | •   | •   | •   |
| Two-piece, Regular Bore    | Fire-safe | Flanged CI 150/ 300      |     |     |     |     |    |       |       | •  | •     | •  | •   | •   | •   |
| Three-piece, Full Bore     | Standard  | Screwed/ Socket-weld*    | •   | •   | •   | •   | •  | •     | •     | •  | •     | •  |     |     |     |
|                            |           | Flanged CI 150/ 300/ 600 |     |     | •   | •   | •  |       | •     | •  |       |    |     |     |     |
|                            |           | Flanged CI 2500          |     |     | •   | •   | •  |       | •     |    |       |    |     |     |     |
|                            | Fire-safe | Screwed/ Socket-weld*    | •   | •   | •   | •   | •  | •     | •     | •  | •     |    |     |     |     |
|                            | IBR       | Screwed/ Socket-weld     | •   | •   | •   | •   | •  | •     | •     | •  |       |    |     |     |     |
| Three-piece, Regular Bore  | Standard  | Screwed/ Socket-weld**   | •   | •   | •   | •   | •  | •     | •     | •  |       | •  | •   |     |     |
|                            |           | Flanged CI 150/ 300/ 600 |     |     |     | •   | •  |       | •     | •  |       |    |     |     |     |
|                            |           | Flanged CI 1500          |     |     | •   | •   | •  |       | •     | •  |       |    |     |     |     |
|                            | Fire-safe | Screwed/ Socket-weld*    |     |     |     | •   | •  | •     | •     | •  |       |    |     |     |     |
|                            | IBR       | Screwed/ Socket-weld     |     |     |     | •   | •  | •     | •     | •  |       |    |     |     |     |

\*Screwed/ socket-weld valves in sizes DN 8 to DN 50 are offered in Class 800, and sizes DN 65 and above, in Class 300.

\*\*Sizes DN 15 to DN 50 are offered in Class 1500 also.

T Ball Valves has a  
 st a matching shoulder  
 owout proof - the  
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**Sealing**  
 further enhanced by  
 n packing rings. To  
 thermal expansions,

**Mirror-finished Solid Stainless Steel Balls**  
 To obtain bubble-tight sealing and lower  
 operating torques, mirror-finished SS balls are

The side-entry stem of L&T Ball Valves has a shoulder that bears against a matching shoulder in the body to make it blowout proof - the higher the line pressure, the better the sealing to atmosphere.

Sealing to atmosphere is further enhanced by stem thrust seals and stem packing rings. To compensate for wear and thermal expansions, Belleville springs are used.

To obtain bubble-tight sealing and lower operating torques, mirror-finished SS balls are used as a standard. The solid construction of the balls guarantees higher structural strength.

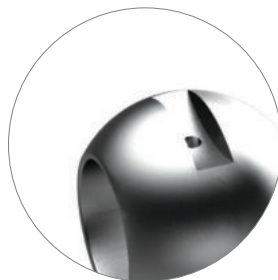
Seats with fine grain structure are used to ensure better strength and longer life. Pressure-relieving slots are a unique feature of L&T Ball Valves. The slots relieve upstream pressure when the valve is in closed position and prevent seat damage.

L&T single-piece and two-piece Ball Valves are provided with an integral actuator mounting flange conforming to ISO 5211.

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### Fire-Safe Feature

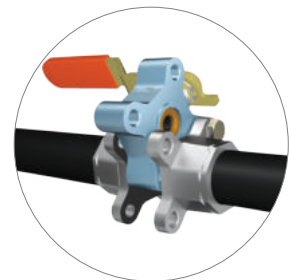
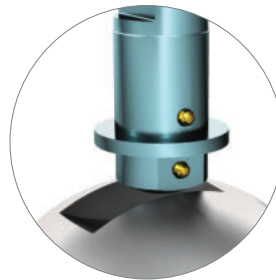
L&T fire-safe Ball Valves feature secondary metal seats. In the event of a fire, after the soft-seat totally sublimates, the ball moves and abuts the downstream metal seat to form a leak-tight seal.



### Antistatic Feature

Static electricity builds up inside a valve when the ball rubs against non-metallic seats. This is a fire hazard, especially in flammable fluid pipelines. L&T Ball Valves have inbuilt antistatic mechanisms to provide electrical continuity.

Stems of full bore valves of size 65 mm and above and regular bore valves of size 80 mm and above are equipped with spring-loaded plungers. In valves of smaller sizes, electrical continuity is achieved by using Carbon-filled PTFE thrust seal and graphite stem packings.



### Cavity Pressure Relief Mechanism

A slight increase in temperature of fluid entrapped in ball cavity can cause a rapid increase of pressure and damage the ball and seats.

To prevent this, L&T Ball Valves are equipped with automatic cavity relief mechanisms. When the valve is in open position, a hole provided on the ball connects the body cavity to the ball port and thereby ensure that cavity pressure does not build up. In the closed condition, when the cavity pressure rises above a designated level, the seat lip deflects to relieve pressure to the upstream side.

### On-line Service

Three-piece Ball Valves are designed for easy on-line service. To access the valve internals, remove three connector bolts and swing the body out using the fourth bolt as a pivot.

## Standards

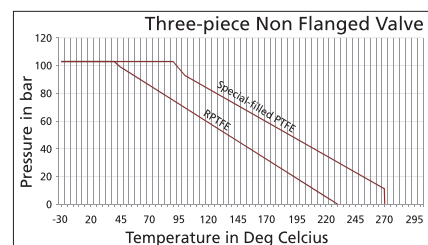
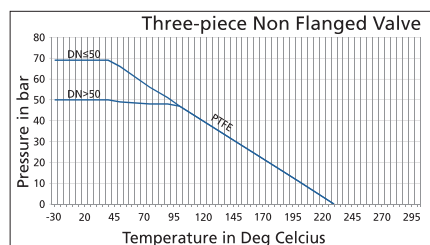
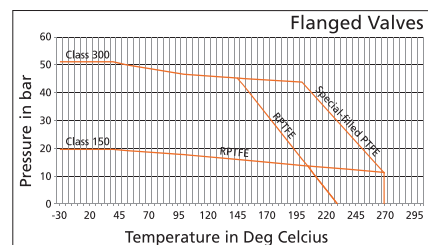
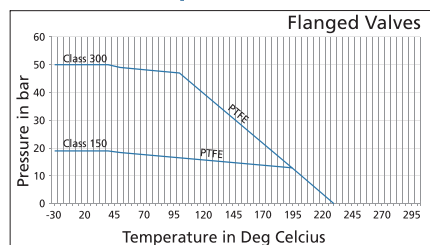
| Scope                  |                  | Standard               |
|------------------------|------------------|------------------------|
| Design*                |                  | ISO 17292/ API 6D      |
| End-to-End             | Socket-weld ends | ASME B16.11            |
|                        | Screwed-end BSPT | ISO 7-1                |
|                        | Screwed-end NPT  | ASME B1.20.1           |
|                        | End flange       | ASME B16.5 RF/ FF/ RTJ |
| Face-to-face (Flanged) |                  | ASME B16.10            |
| Pressure testing       |                  | ISO 5208/ API 598      |
| Fire Test              |                  | ISO 10497/ API 607     |

\*Valve with API 6D design can be offered on demand

## Material of Construction

| No. | Part                          | Material specification                                              |           |                                                  |
|-----|-------------------------------|---------------------------------------------------------------------|-----------|--------------------------------------------------|
|     |                               | Single-piece                                                        | Two-piece | Three-piece                                      |
| 1   | Body/<br>Connector/<br>Insert | ASTM A216 Gr. WCB                                                   |           | ASTM A216 Gr. WCB<br>ASTM A105                   |
|     |                               | ASTM A351 Gr. CF8M/ ASTM A182 Gr. F316                              |           |                                                  |
| 2   | Ball                          | ASTM A351 Gr. CF8M/ ASTM A182 Gr. F316/ ASTM A182 Gr. F304          |           |                                                  |
| 3   | Seat                          | PTFE/ RPTFE/ Special filled PTFE/ PEEK/ Devlon®/ NXT-70/ SS316 + ST |           |                                                  |
| 4   | Stem                          | ASTM A479 Type 316                                                  |           |                                                  |
| 5   | Body Seal                     | Graphite                                                            |           | PTFE/ RPTFE/<br>Special filled PTFE/<br>Graphite |
| 6   | Stem Thrust Seal              | Carbon-filled PTFE                                                  |           |                                                  |
| 7   | Stem Packing                  | Carbon-filled PTFE/ Graphite                                        |           |                                                  |

## Pressure Temperature Charts - Seat Materials



## Test Pressures (bar)

|                     | Class 150 | Class 300 | Class 600* |
|---------------------|-----------|-----------|------------|
| Shell - Hydrostatic | 30        | 77        | 155        |
| Seat - Hydrostatic  | 22        | 56        | 113        |
| Seat - Pneumatic    | 5.5       | 5.5       | 5.5        |

\*With RPTFE/ Special filled PTFE

## Valve Torque Data (in Nm)

| Size |     | Bore | Single-piece      |                   | Two-piece         |                   | Three-piece                 |                   |                   |
|------|-----|------|-------------------|-------------------|-------------------|-------------------|-----------------------------|-------------------|-------------------|
| DN   | NPS |      | Flanged<br>Cl 150 | Flanged<br>Cl 300 | Flanged<br>Cl 150 | Flanged<br>Cl 300 | Screwed/<br>Socket-<br>weld | Flanged<br>Cl 150 | Flanged<br>Cl 300 |
| 8    | ¼   | FB   |                   |                   |                   |                   | 6.5                         |                   |                   |
|      |     | RB   |                   |                   |                   |                   |                             |                   |                   |
| 10   | ⅜   | FB   |                   |                   |                   |                   | 6.5                         |                   |                   |
|      |     | RB   |                   |                   |                   |                   |                             |                   |                   |
| 15   | ½   | FB   | 5                 | 5                 | 9                 | 9                 | 6.5                         | 5                 | 5                 |
|      |     | RB   |                   |                   |                   |                   |                             |                   |                   |
| 20   | ¾   | FB   |                   |                   | 9                 | 9                 | 9                           | 8                 | 9                 |
|      |     | RB   | 5                 | 5                 |                   |                   | 6.5                         | 5                 | 5                 |
| 25   | 1   | FB   |                   |                   | 14                | 16                | 11                          | 10                | 11                |
|      |     | RB   | 12                | 15                |                   |                   | 9                           | 8                 | 9                 |
| 32   | 1¼  | FB   |                   |                   |                   |                   | 30                          |                   |                   |
|      |     | RB   |                   |                   |                   |                   | 16                          |                   |                   |
| 40   | 1½  | FB   |                   |                   | 25                | 28                | 34                          | 18                | 32                |
|      |     | RB   | 18                | 22                |                   |                   | 13                          | 13                | 13                |
| 50   | 2   | FB   |                   |                   | 65                | 70                | 44                          | 26                | 30                |
|      |     | RB   | 38                | 40                |                   |                   | 34                          | 30                | 32                |
| 65   | 2½  | FB   |                   |                   | 90                | 98                |                             |                   |                   |
|      |     | RB   | 48                | 60                |                   |                   |                             |                   |                   |
| 80   | 3   | FB   |                   |                   | 150               | 165               |                             |                   |                   |
|      |     | RB   | 90                | 100               |                   |                   |                             |                   |                   |
| 100  | 4   | FB   |                   |                   | 165               | 185               |                             |                   |                   |
|      |     | RB   | 145               | 160               |                   |                   |                             |                   |                   |
| 150  | 6   | FB   |                   |                   | 250               | 350               |                             |                   |                   |
|      |     | RB   | 165               | 190               |                   |                   |                             |                   |                   |
| 200  | 8   | FB   |                   |                   | 650               | 750               |                             |                   |                   |
|      |     | RB   |                   |                   |                   |                   |                             |                   |                   |

For screwed/ socket-weld end three-piece valves, torque values are as per full-rated working pressure of 69 bar. Indicated design torque values are without factor of safety. Torque values for other sizes/ classes will be provided on demand.

## Flow Co-efficients

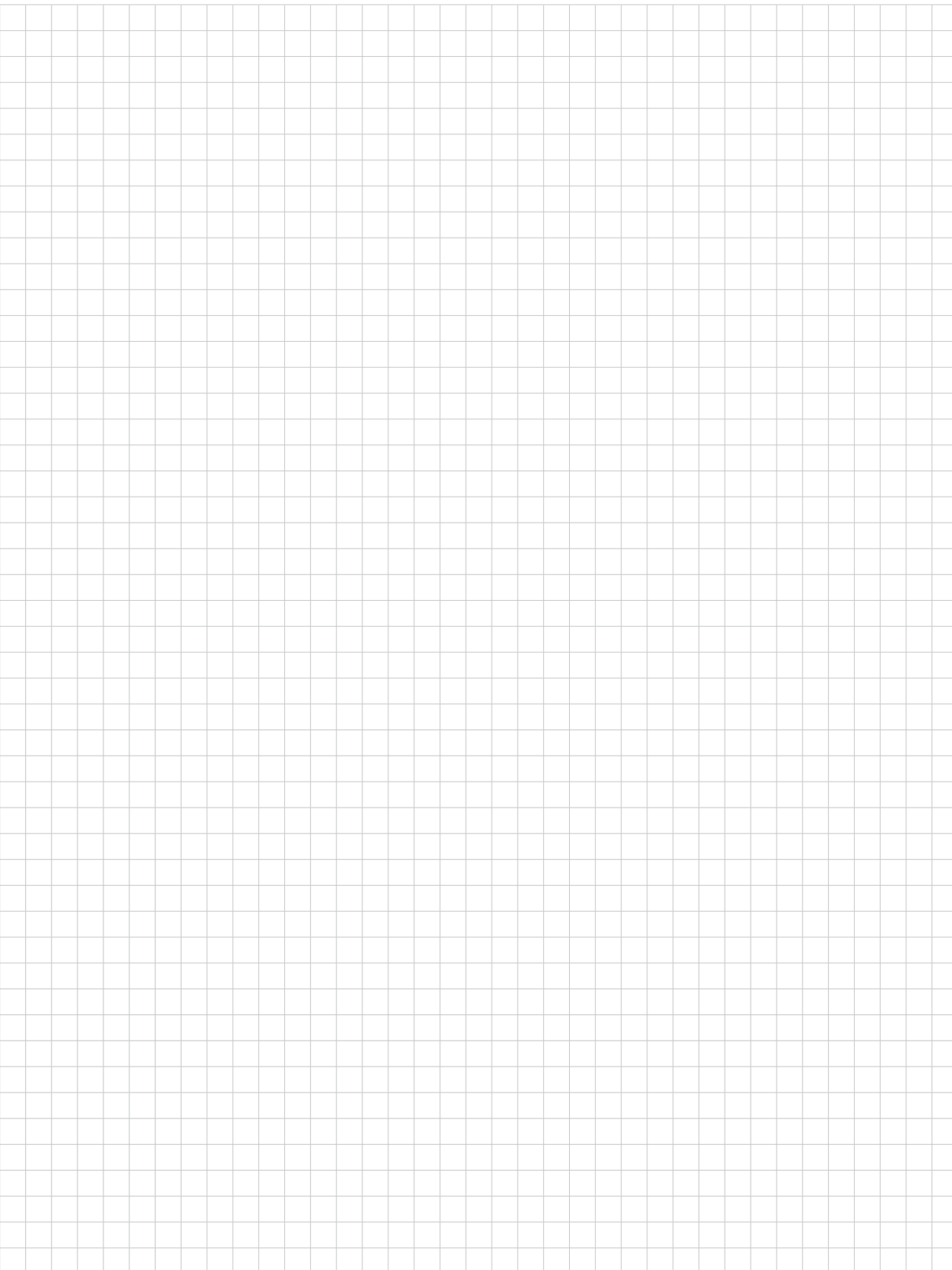
|                | Bore | 8 | 10 | 15 | 20 | 25  | 32  | 40  | 50  | 65  | 80   | 100  | 150  | 200  |
|----------------|------|---|----|----|----|-----|-----|-----|-----|-----|------|------|------|------|
| C <sub>v</sub> | FB   | 8 | 9  | 20 | 50 | 100 | 135 | 225 | 465 | 770 | 1090 | 1970 | 4635 | 8565 |
|                | RB   |   |    |    | 15 | 40  | 60  | 95  | 175 | 335 | 570  | 625  | 870  | 2300 |
| K <sub>v</sub> | FB   | 7 | 8  | 17 | 45 | 85  | 115 | 220 | 395 | 655 | 930  | 1675 | 3940 | 7280 |
|                | RB   |   |    |    | 13 | 35  | 50  | 80  | 150 | 285 | 485  | 530  | 740  | 1955 |

C<sub>v</sub> - Flow Co-efficient of a valve is defined as flow of water at 15.6°C (60°F) in gallon (US) per minute at a pressure drop of 1 psi across the valve. K<sub>v</sub> - Flow Co-efficient of a valve is defined as flow of water with temperature ranging from 5°C (40°F) to 40°C (104°F) in cubic meter per hour (m³/hr) at a pressure drop of 1 bar across the valve. C<sub>v</sub> and K<sub>v</sub> values are given for valve in fully open condition.

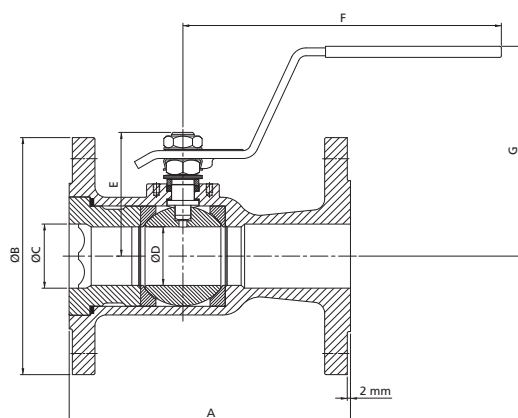


## End Connection Identification

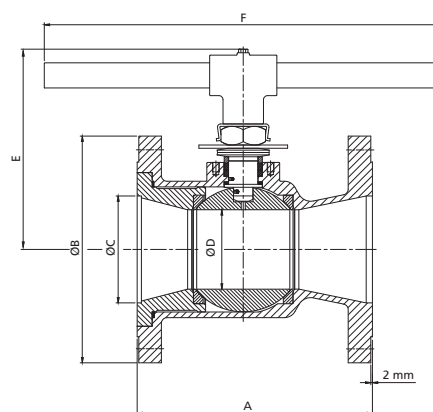
Grooves are provided on body connectors to differentiate valves based on end connections - one groove for socket-weld ends, two grooves for valves with screwed ends (NPT threads) and no grooves for valves with screwed ends (BSPT threads).



## Single-piece Ball Valves - Regular Bore (ASME Class 150 & 300)



Lever operated  
(DN 15 - DN 65)



Wrench operated  
(DN 80 - DN 150)

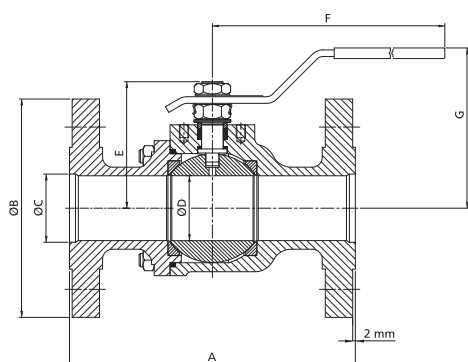
### Dimensions

| Size |     | A      |        | B      |        | C   | D  | E   | F   | G   | Wt.    |        | ISO 5211 Base |        |
|------|-----|--------|--------|--------|--------|-----|----|-----|-----|-----|--------|--------|---------------|--------|
| DN   | NPS | CI 150 | CI 300 | CI 150 | CI 300 |     |    |     |     |     | CI 150 | CI 300 | CI 150        | CI 300 |
| 15   | ½   | 108    | 140    | 90     | 95     | 13  | 11 | 38  | 120 | 90  | 1      | 2      | F03           | F03    |
| 20   | ¾   | 117    | 152    | 100    | 115    | 19  | 11 | 38  | 120 | 90  | 1      | 2.5    | F03           | F03    |
| 25   | 1   | 127    | 165    | 110    | 125    | 25  | 17 | 46  | 140 | 100 | 2      | 4      | F03           | F03    |
| 40   | 1½  | 165    | 190    | 125    | 155    | 38  | 27 | 56  | 180 | 115 | 4      | 4.5    | F04           | F04    |
| 50   | 2   | 178    | 216    | 150    | 165    | 51  | 37 | 75  | 210 | 122 | 6.5    | 9      | F05           | F05    |
| 65   | 2½  | 190    | 241    | 180    | 190    | 64  | 49 | 102 | 210 | 142 | 11     | 15     | F07           | F07    |
| 80   | 3   | 203    | 282    | 190    | 210    | 76  | 62 | 175 | 390 |     | 15.5   | 22     | F07           | F07    |
| 100  | 4   | 229    | 305    | 230    | 255    | 102 | 74 | 195 | 390 |     | 25     | 35     | F07           | F07    |
| 150  | 6   | 267    | 403    | 280    | 320    | 152 | 98 | 245 | *   |     | 40     | 65     | F10           | F12    |

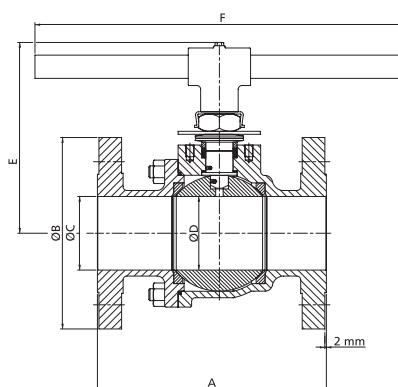
All dimensions in mm and weights in kg

\*600 mm for CI 150 & 890 mm for CI 300

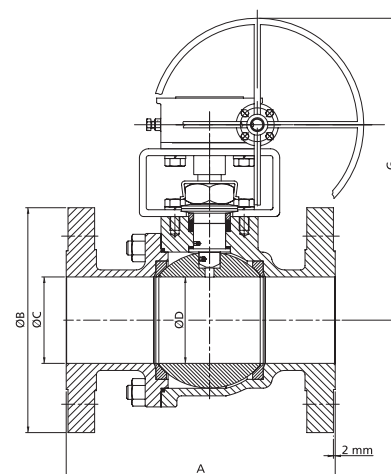
## Two-piece Ball Valves - Full Bore (ASME Class 150 & 300)



Lever operated  
(DN 15 - DN 50)



Wrench operated  
(DN 65 - DN 200)



Gear operated  
(DN 100 - DN 200)

### Dimensions

| Size |     | A      |        | B      |        | C   | D   | E      |        | F   | G   | Wt.    |        | ISO 5211 Base |        |
|------|-----|--------|--------|--------|--------|-----|-----|--------|--------|-----|-----|--------|--------|---------------|--------|
| DN   | NPS | CI 150 | CI 300 | CI 150 | CI 300 |     |     | CI 150 | CI 300 |     |     | CI 150 | CI 300 | CI 150        | CI 300 |
| 15   | ½   | 108    | 140    | 90     | 95     | 13  | 11  | 38     | 38     | 120 | 90  | 1      | 2      |               |        |
| 20   | ¾   | 117    | 152    | 100    | 115    | 19  | 17  | 45     | 45     | 120 | 97  | 2      | 3      |               |        |
| 25   | 1   | 127    | 165    | 110    | 125    | 25  | 24  | 55     | 55     | 158 | 97  | 3      | 4      |               |        |
| 40   | 1½  | 165    | 190    | 125    | 155    | 38  | 37  | 75     | 75     | 212 | 122 | 5      | 6      | F05           | F05    |
| 50   | 2   | 178    | 216    | 150    | 165    | 51  | 49  | 103    | 103    | 212 | 142 | 9      | 12     | F07           | F07    |
| 65   | 2½  | 190    | 241    | 180    | 190    | 64  | 62  | 175    | 175    | 390 |     | 14     | 20.5   | F07           | F07    |
| 80   | 3   | 203    | 282    | 190    | 210    | 76  | 74  | 185    | 195    | 390 |     | 19     | 28     | F07           | F07    |
| 100  | 4   | 229    | 305    | 230    | 255    | 102 | 98  | 235    | 247    | *   |     | 31     | 45     | F10           | F12    |
| 150  | 6   | 394    | 403    | 280    | 320    | 152 | 148 | 285    | 315    | **  |     | 78     | 105    | F12           | F14    |
| 200  | 8   | 457    | 502    | 345    | 380    | 203 | 198 | 345    | 371    | 980 |     | 141.5  | 169    | F14           | F16    |

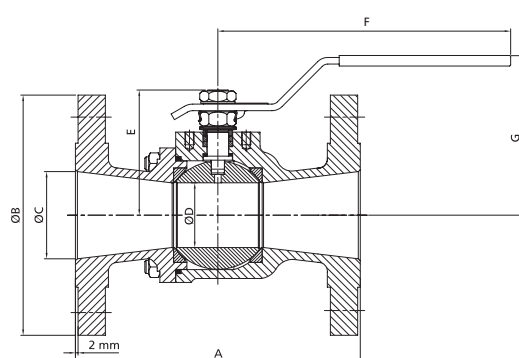
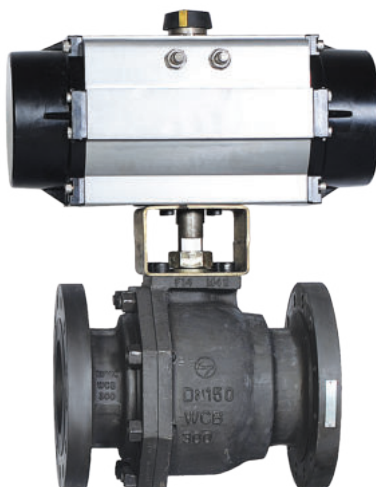
All dimensions in mm and weights in kg

\*600 mm for CI 150 & 890 mm for CI 300

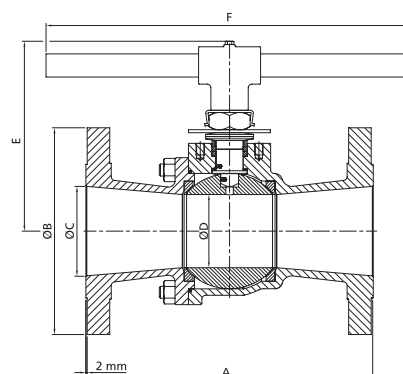
\*\*890 mm for CI 150 & 980 mm for CI 300

API 6D design available on demand. Also metal-seated valves in sizes up to DN 80.

## Two-piece Ball Valves - Regular Bore (ASME Class 150 & 300)



Lever operated  
(DN 50 & DN 65)



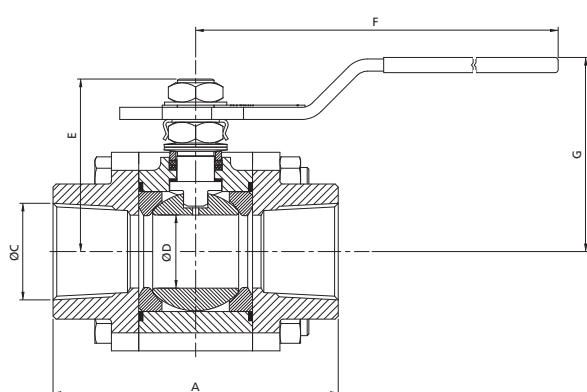
Wrench operated  
(DN 80 - DN 200)

### Dimensions

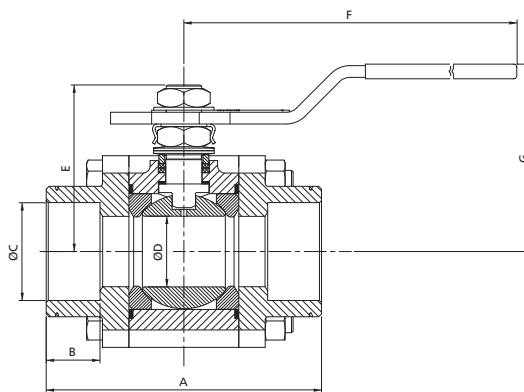
| Size |     | A      |        | B      |        | C   | D   | E      |        | F   | G   | Wt.    |        | ISO 5211 Base |        |
|------|-----|--------|--------|--------|--------|-----|-----|--------|--------|-----|-----|--------|--------|---------------|--------|
| DN   | NPS | CI 150 | CI 300 | CI 150 | CI 300 |     |     | CI 150 | CI 300 |     |     | CI 150 | CI 300 | CI 150        | CI 300 |
| 50   | 2   | 178    | 216    | 150    | 165    | 51  | 37  | 75     | 75     | 212 | 122 | 7      | 10     | F05           | F05    |
| 65   | 2½  | 190    | 241    | 180    | 190    | 64  | 49  | 103    | 103    | 212 | 142 | 12     | 17     | F07           | F07    |
| 80   | 3   | 203    | 282    | 190    | 210    | 76  | 62  | 175    | 175    | 390 |     | 20     | 24     | F07           | F07    |
| 100  | 4   | 229    | 305    | 230    | 255    | 102 | 74  | 185    | 195    | 390 |     | 28     | 45     | F07           | F07    |
| 150  | 6   | 394    | 403    | 280    | 320    | 152 | 98  | 235    | 247    | 600 |     | 52     | 5      | F10           | F12    |
| 200  | 8   | 457    | 502    | 345    | 380    | 203 | 148 | 285    | 315    | 890 |     | 95     | 142    | F12           | F14    |

All dimensions in mm and weights in kg

## Three-piece Ball Valves - Full Bore (ASME Class 150, 300, 600 & 2500)



Screwed



Socket-weld end

### Dimensions

Screwed/ Socket-weld end

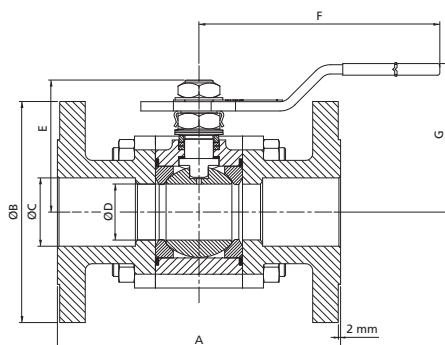
| Size |     | A   | B   | C             |          |           | D  | E   | F   | G   | Wt.  |
|------|-----|-----|-----|---------------|----------|-----------|----|-----|-----|-----|------|
| DN   | NPS |     |     | SW            | BSPT     | NPT       |    |     |     |     |      |
| 8    | ¼   | 62  | 9.5 | 14.6 - 14.2   | ¼" - 19  | ¼" - 18   | 11 | 38  | 132 | 50  | 1    |
| 10   | ⅜   | 62  | 9.5 | 18.0 - 17.6   | ⅜" - 19  | ⅜" - 18   | 11 | 38  | 132 | 50  | 1    |
| 15   | ½   | 66  | 10  | 22.2 - 21.8   | ½" - 14  | ½" - 14   | 11 | 38  | 132 | 50  | 1    |
| 20   | ¾   | 75  | 13  | 27.6 - 27.2   | ¾" - 14  | ¾" - 14   | 17 | 46  | 132 | 58  | 1    |
| 25   | 1   | 92  | 13  | 34.3 - 33.9   | 1" - 11  | 1" - 11½  | 24 | 54  | 162 | 65  | 2    |
| 32   | 1¼  | 114 | 13  | 43.1 - 42.7   | 1¼" - 11 | 1¼" - 11½ | 37 | 75  | 192 | 83  | 4    |
| 40   | 1½  | 114 | 13  | 49.2 - 48.8   | 1½" - 11 | 1½" - 11½ | 37 | 75  | 192 | 83  | 4    |
| 50   | 2   | 132 | 16  | 61.7 - 61.2   | 2" - 11  | 2" - 11½  | 49 | 87  | 202 | 94  | 6    |
| 65   | 2½  | 185 | 16  | 74.4 - 73.9   | 2½" - 11 | 2½" - 8   | 62 | 200 | 390 | 200 | 15.5 |
| 80   | 3   | 208 | 16  | 90.3 - 89.8   | 3" - 11  | 3" - 8    | 74 | 225 | 390 | 225 | 25   |
| 100  | 4   | 240 | 19  | 115.7 - 115.2 | 4" - 11  | 4" - 8    | 98 | 250 | 600 | 250 | 39   |

Cold Working Pressure (CWP)

- 69 bar, with PTFE seat
- 103 bar, with RPTFE and special filled PTFE
- 400 bar, with Devlon®
- 425 bar, with PEEK

All dimensions in mm and weights in kg

Screwed/ socket-weld valves in sizes DN 8 to DN 50 have a body rating of Class 800. In larger sizes, the body is rated to Class 300.

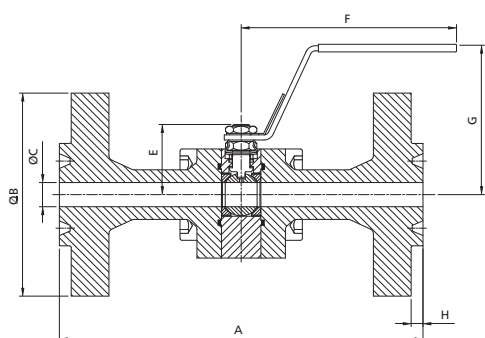


## Dimensions

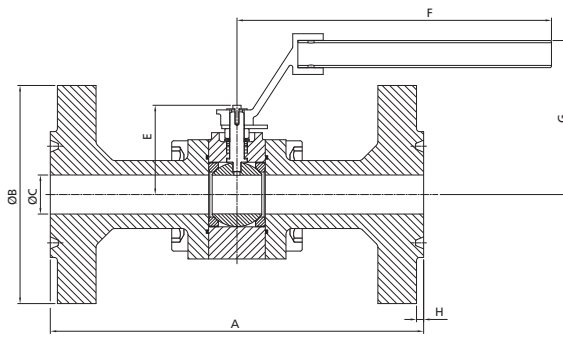
Flanged end - Class 150, 300 & 600

| Size |     | A      |        |        | B      |        |        | C  | D  | E  | F   | G   | Wt.    |        |        |
|------|-----|--------|--------|--------|--------|--------|--------|----|----|----|-----|-----|--------|--------|--------|
| DN   | NPS | CI 150 | CI 300 | CI 600 | CI 150 | CI 300 | CI 600 |    |    |    |     |     | CI 150 | CI 300 | CI 600 |
| 15   | ½   | 108    | 140    | 165    | 90     | 95     | 95     | 13 | 11 | 38 | 122 | 92  | 2      | 2      | 2.5    |
| 20   | ¾   | 117    | 152    | 190    | 100    | 115    | 115    | 19 | 17 | 46 | 142 | 98  | 2      | 3      | 3.5    |
| 25   | 1   | 127    | 165    | 216    | 110    | 125    | 125    | 25 | 24 | 54 | 152 | 98  | 3      | 5      | 6.5    |
| 40   | 1½  | 165    | 190    | 241    | 125    | 155    | 155    | 38 | 37 | 75 | 212 | 125 | 8      | 11     | 12     |
| 50   | 2   | 178    | 216    | 292    | 150    | 165    | 165    | 51 | 49 | 87 | 212 | 125 | 13     | 15.5   | 12.5   |

All dimensions in mm and weights in kg



Lever operated  
(DN 15 - DN 25)



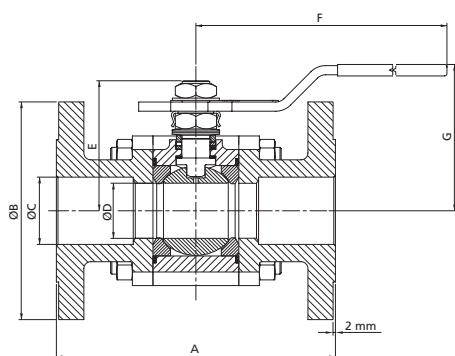
Wrench operated  
(DN 40)

Flanged end - Class 2500

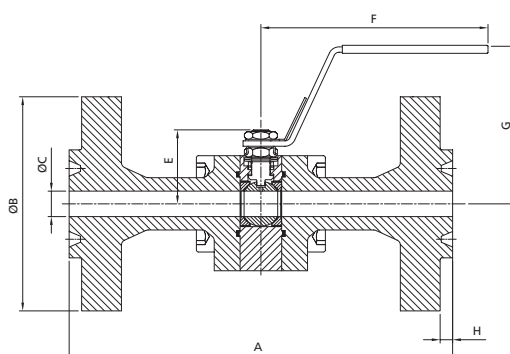
| Size |     | A   | B   | C    | E   | F   | G   | H    | Wt. |
|------|-----|-----|-----|------|-----|-----|-----|------|-----|
| DN   | NPS |     |     |      |     |     |     |      |     |
| 15   | ½   | 264 | 135 | 14.3 | 50  | 195 | 103 | 6.35 | 10  |
| 20   | ¾   | 273 | 140 | 20.6 | 66  | 250 | 125 | 6.35 | 14  |
| 25   | 1   | 308 | 160 | 25.3 | 92  | 360 | 153 | 6.35 | 25  |
| 40   | 1½  | 386 | 205 | 38.5 | 100 | 315 | 162 | 7.92 | 40  |

All dimensions in mm and weights in kg

## Three-piece Ball Valves - Regular Bore (ASME Class 150, 300, 600 & 1500)



Class 150, 300 & 600



Class 1500

### Dimensions

Flanged end - Class 150, 300 & 600

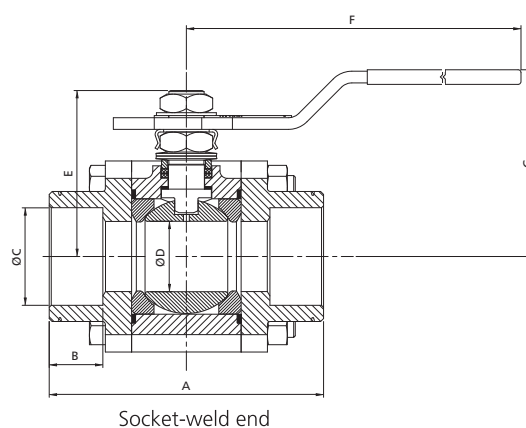
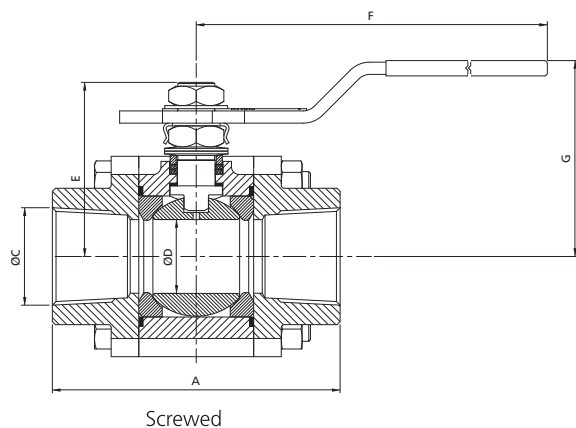
| Size |     | A      |        |        | B      |        |        | C  | D  | E  | F   | G   | Wt.    |        |        |
|------|-----|--------|--------|--------|--------|--------|--------|----|----|----|-----|-----|--------|--------|--------|
| DN   | NPS | CI 150 | CI 300 | CI 600 | CI 150 | CI 300 | CI 600 |    |    |    |     |     | CI 150 | CI 300 | CI 600 |
| 20   | ¾   | 117    | 152    | 190    | 100    | 115    | 115    | 19 | 11 | 38 | 122 | 92  | 1.8    | 3.0    | 4      |
| 25   | 1   | 127    | 165    | 216    | 110    | 125    | 125    | 25 | 17 | 46 | 142 | 98  | 2.5    | 4.2    | 5.5    |
| 40   | 1½  | 165    | 190    | 241    | 125    | 155    | 155    | 38 | 27 | 56 | 180 | 117 | 4.7    | 8.0    | 9.5    |
| 50   | 2   | 178    | 216    | 292    | 150    | 165    | 165    | 51 | 37 | 75 | 212 | 125 | 7.7    | 12.1   | 10.5   |

All dimensions in mm and weights in kg

Flanged end - Class 1500

| Size |     | A   | B   | C    | E  | F   | G   | H    | Wt. |
|------|-----|-----|-----|------|----|-----|-----|------|-----|
| DN   | NPS |     |     |      |    |     |     |      |     |
| 15   | ½   | 216 | 120 | 14.3 | 41 | 120 | 88  | 6.35 | 7   |
| 20   | ¾   | 229 | 130 | 14.3 | 41 | 120 | 88  | 6.35 | 8   |
| 25   | 1   | 254 | 150 | 20.6 | 54 | 235 | 120 | 6.35 | 10  |
| 40   | 1½  | 305 | 180 | 25.4 | 54 | 235 | 127 | 6.35 | 18  |
| 50   | 2   | 368 | 215 | 38.5 | 93 | 450 | 157 | 7.92 | 27  |

All dimensions in mm and weights in kg



## Dimensions

### Screwed/ Socket-weld end

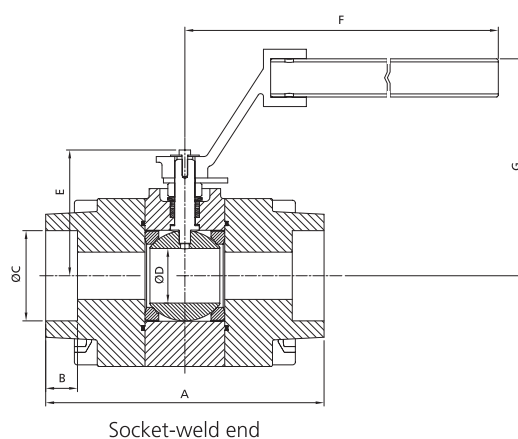
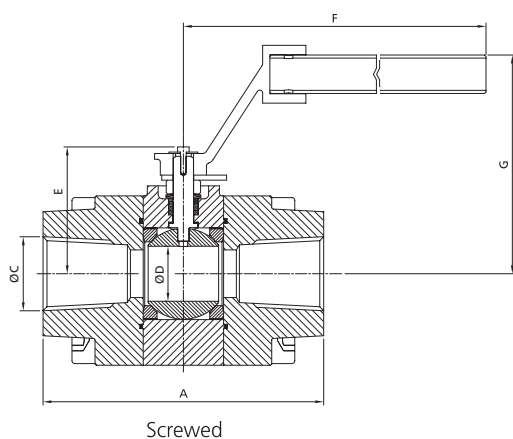
| Size |     | A   | B   | C             |          |           | D  | E   | F   | G   | Wt. |
|------|-----|-----|-----|---------------|----------|-----------|----|-----|-----|-----|-----|
| DN   | NPS |     |     | SW            | BSPT     | NPT       |    |     |     |     |     |
| 8    | ¼   | 62  | 9.5 | 14.6 - 14.2   | ¼" - 19  | ¼" - 18   | 11 | 38  | 132 | 50  | 1   |
| 10   | ⅜   | 62  | 9.5 | 18.0 - 17.6   | ⅜" - 19  | ¾" - 18   | 11 | 38  | 132 | 50  | 1   |
| 15   | ½   | 66  | 10  | 22.2 - 21.8   | ½" - 14  | ½" - 14   | 11 | 38  | 132 | 50  | 1   |
| 20   | ¾   | 69  | 13  | 27.6 - 27.2   | ¾" - 14  | ¾" - 14   | 11 | 38  | 132 | 50  | 1   |
| 25   | 1   | 90  | 13  | 34.3 - 33.9   | 1" - 11  | 1" - 11½  | 17 | 46  | 132 | 58  | 1   |
| 32   | 1¼  | 97  | 13  | 43.1 - 42.7   | 1¼" - 11 | 1¼" - 11½ | 24 | 54  | 162 | 65  | 2   |
| 40   | 1½  | 103 | 13  | 49.2 - 48.8   | 1½" - 11 | 1½" - 11½ | 27 | 56  | 162 | 66  | 2.5 |
| 50   | 2   | 122 | 16  | 61.7 - 61.2   | 2" - 11  | 2" - 11½  | 37 | 75  | 192 | 83  | 4.5 |
| 80   | 3   | 185 | 16  | 90.3 - 89.8   | 3" - 11  | 3" - 8    | 62 | 200 | 390 | 200 | 17  |
| 100  | 4   | 212 | 19  | 115.7 - 115.2 | 4" - 11  | 4" - 8    | 74 | 225 | 600 | 225 | 27  |

Cold Working Pressure (CWP)

- 69 bar, with PTFE seat
- 103 bar, with RPTFE and special filled PTFE
- 400 bar, with Devlon®
- 425 bar, with PEEK

All dimensions in mm and weights in kg

Valves in sizes DN 8 to DN 15 are offered in full bore construction



### Screwed/ Socket-weld end - Class 1500

| Size |     | A   | B           | C           |          |           | D    | E  | F   | G   | Wt. |
|------|-----|-----|-------------|-------------|----------|-----------|------|----|-----|-----|-----|
| DN   | NPS |     |             | SW          | BSPT     | NPT       |      |    |     |     |     |
| 15   | ½   | 85  | 10.5 - 10.0 | 22.2 - 21.8 | ½" - 14  | ½" - 14   | 14.3 | 41 | 120 | 52  | 3   |
| 20   | ¾   | 85  | 13.5 - 13.0 | 27.6 - 27.2 | ¾" - 14  | ¾" - 14   | 14.3 | 41 | 120 | 52  | 3.5 |
| 25   | 1   | 108 | 13.5 - 13.0 | 34.3 - 33.9 | 1" - 11  | 1" - 11½  | 20.6 | 54 | 130 | 90  | 4   |
| 40   | 1½  | 125 | 13.5 - 13.0 | 49.2 - 48.8 | 1½" - 11 | 1½" - 11½ | 25.4 | 54 | 130 | 90  | 5   |
| 50   | 2   | 142 | 16.5 - 16.0 | 61.7 - 61.2 | 2" - 11  | 2" - 11½  | 38.5 | 88 | 210 | 155 | 6   |

All dimensions in mm and weights in kg

## Ordering Information

| Series | Type                                | Bore        | Ends*                                | Material            | Option        | Operator         |
|--------|-------------------------------------|-------------|--------------------------------------|---------------------|---------------|------------------|
| L      | 1 - Single-piece                    | F - Full    | BT - BSPT Threaded                   | C - Carbon Steel    | F - Fire-safe | Lever/<br>Wrench |
|        | 2 - Two-piece                       | R - Regular | NT - NPT Threaded                    | S - Stainless Steel |               | Gear Unit        |
|        | 3 - Three-piece                     |             | SW - Socket-weld                     | D - Duplex SS       |               | Pneumatic        |
|        | 4 - Two-piece<br>(API 6D)           |             | SN - SW NPT                          |                     |               | Electrical       |
|        | 5 - Two-piece (IBR)                 |             | F1 - Flanged CI 150                  |                     |               |                  |
|        | 6 - Three-piece (IBR)               |             | F3 - Flanged CI 300                  |                     |               |                  |
|        | 8 - Three-piece -<br>Pharmaceutical |             | F4 - Flanged CI 150<br>Flat Face     |                     |               |                  |
|        |                                     |             | F5 - Flanged CI 300<br>Flat Face     |                     |               |                  |
|        |                                     |             | F6 - Flanged CI 600<br>(RF 125-250)  |                     |               |                  |
|        |                                     |             | F9 - Flanged CI 1500<br>(RF 125-250) |                     |               |                  |
|        |                                     |             | FA - Flanged CI 1500<br>(RTJ)        |                     |               |                  |
|        |                                     |             | FR - Flanged CI 2500<br>(RF 125-250) |                     |               |                  |
|        |                                     |             | FC - Flanged CI 2500<br>(RTJ)        |                     |               |                  |

\*Valves with other ends  
available on demand



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